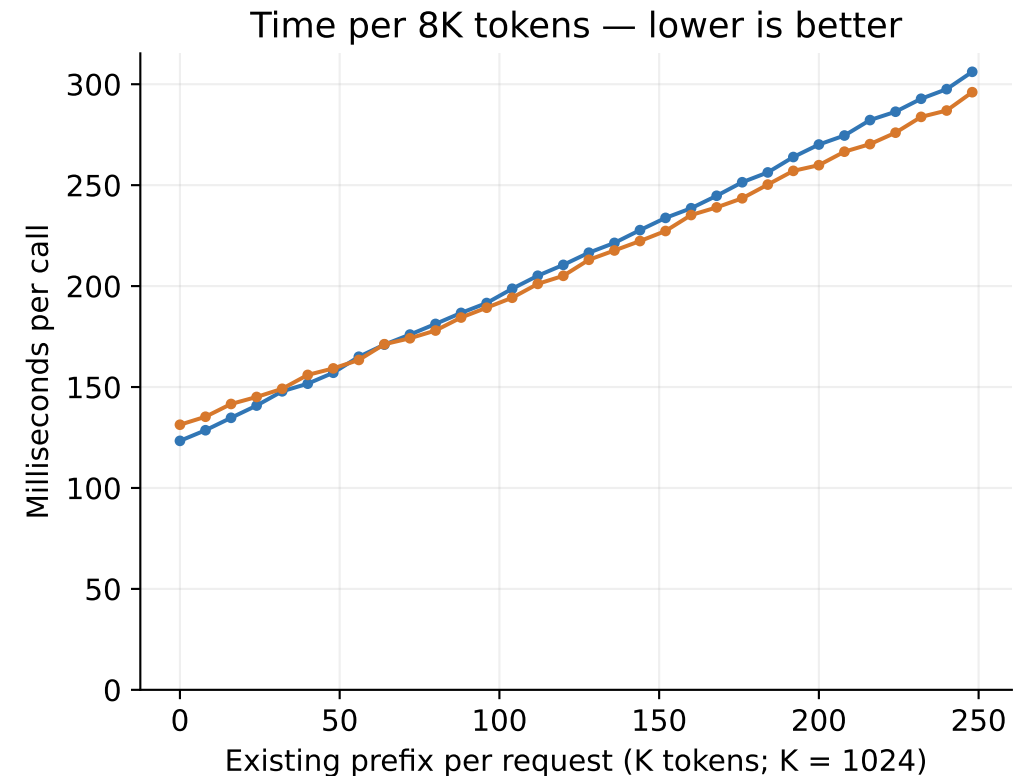
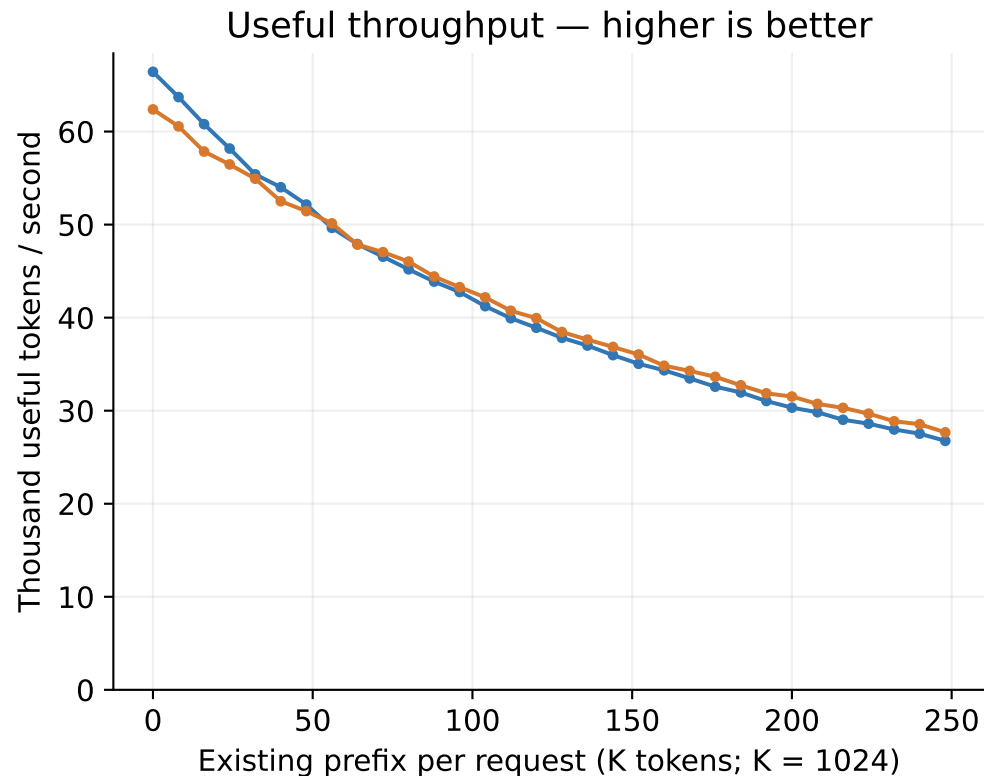


Fixed 2×4K batching vs canonical 8K

Same token count per call: 8,192 useful tokens. One request × 8K versus 2 requests × 4K.

— Canonical: 1 request × 8K — Fixed batch: 2 requests × 4K



0K prefix: canonical 123.3 ms; fixed batch 131.3 ms. The batch takes 6.5% longer for the same token count.

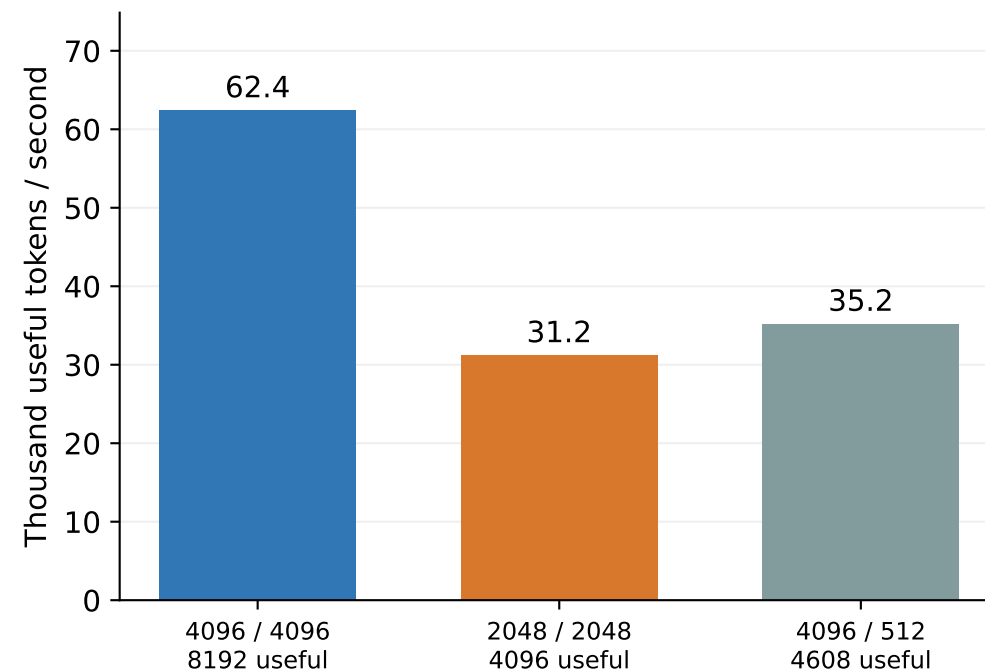
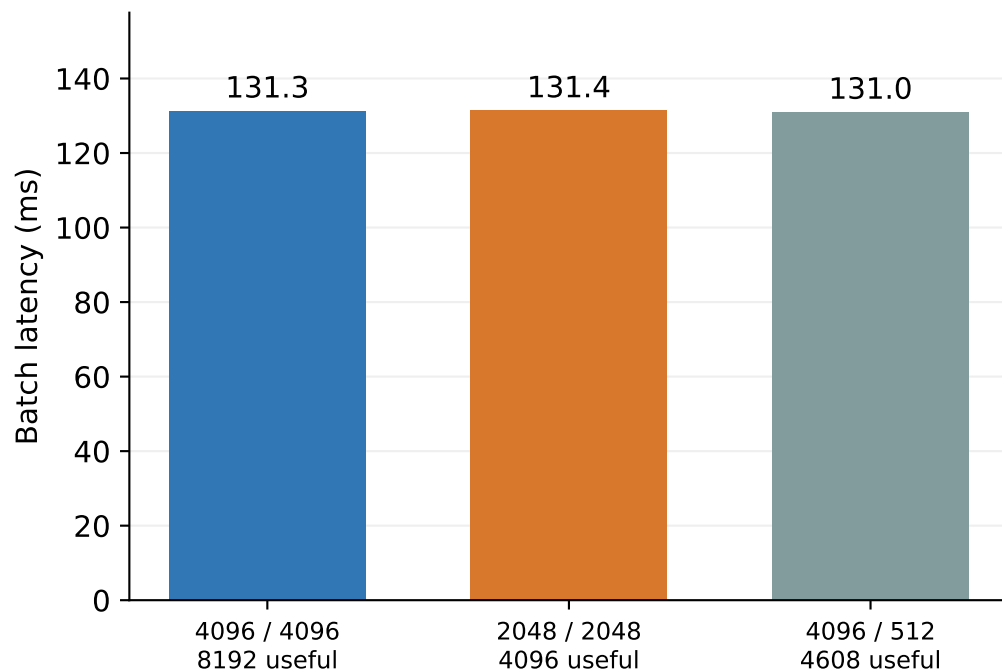
128K prefix: canonical 216.5 ms; fixed batch 213.0 ms. The batch takes 1.6% less time for the same token count.

248K prefix: canonical 306.2 ms; fixed batch 296.1 ms. The batch takes 3.3% less time for the same token count.

Gemma-4-31B-it, 60 layers, Blackhole CP8/TP4, original reduce-scatter, default activation placement. Each call processes 8,192 useful tokens: 1×8K in canonical, or 2×4K in the fixed batch. Every request has the prefix shown on the chart. All histories were populated by real model calls. Five replays at selected prefix positions; one replay elsewhere. JSON records contain every sample and repeat count. Timing covers trace replay and synchronization. Host staging is recorded separately; loading, compilation, output download and KV migration are excluded.

What does fixed-size padding cost?

All cases execute the same $2 \times 4K$ shape. Short final requests still occupy their full 4K lane.



Why measure this? A fixed batch always computes 4K rows per request when fewer tokens are useful. All three cases start at zero and reuse the same trace.

The model shares embedding, projections, norms, RoPE and MLP across 2 requests. Each attention layer runs 2 sequential 4K attention calls with local slices and one local concatenation. It adds no CP all-gathers.

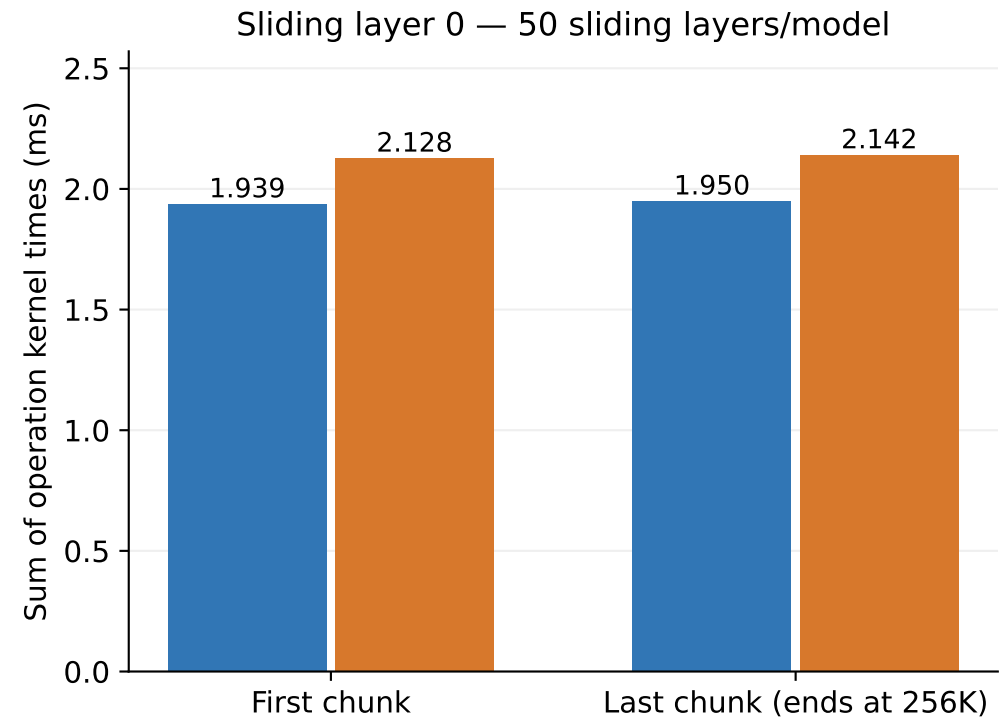
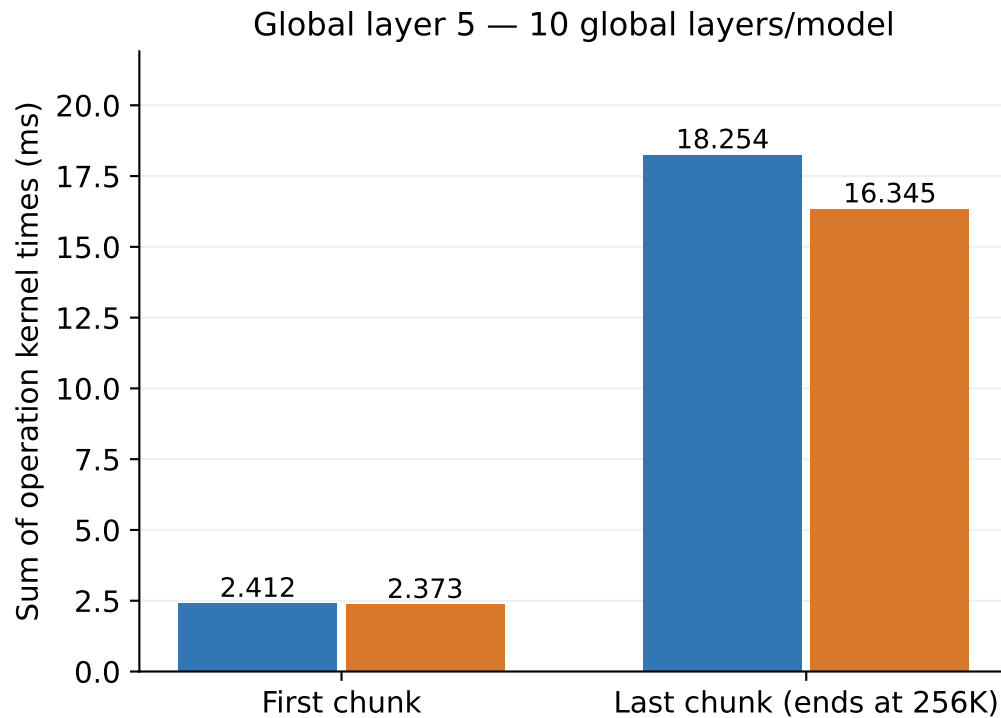
Memory: 2 full 256K request caches use 7.06 GiB/device; canonical uses 3.53 GiB/device. Weights and other buffers are additional. KV storage scales with the number of request slots.

Validation: Six-layer $2 \times 4K$ output/KV, isolation, migration and trace checks passed (72 output/KV comparisons). Original math settings retained.

Where does batching change time?

Isolated layer profiles. Both paths process 8,192 tokens; lower bars are better.

Canonical 1×8K Batch 2×4K



Global, last: batching saves 1.910 ms per layer. The 2 attention calls total 14.364 ms versus 16.363 ms for canonical's one call, Attention changes by -1.998 ms.

Sliding, last: batching adds 0.191 ms per layer. The 2 attention calls total 0.474 ms versus 0.383 ms for canonical's one call, Attention changes by +0.091 ms.

Prefix control: batched global time is 15.974 ms at 248K and 16.345 ms at 252K. This isolates the effect of the different final-chunk starts.

Token ranges (K = 1024): first = canonical [0, 8K), batch 2 × [0, 4K). Last = canonical [248K, 256K), batch 2 × [252K, 256K).

Kernel sums use isolated layers with random KV histories; full-model charts use actual histories. The 50 sliding layers offset much of the saving across 10 global layers. Each panel has its own vertical scale.

Global layer: operation comparison

All times are microseconds, summed over the calls in one layer. C = Canonical 1×8K; B = Batch 2×4K.

Operation	Calls C	Calls B	First C μ s	First B μ s	Extra μ s	Last C μ s	Last B μ s	Extra μ s
Norm input redistribution	4	4	27.2	27.4	+0.2	27.0	26.9	-0.0
RMS normalization	6	6	74.5	74.9	+0.4	74.6	74.3	-0.3
Norm output redistribution	4	4	30.4	30.6	+0.2	30.4	30.5	+0.0
TP all-gather	2	2	212.2	211.4	-0.8	203.0	197.4	-5.6
Q/K/V projection	1	1	160.0	160.2	+0.2	158.8	157.6	-1.2
Split Q/K/V heads	1	1	38.7	39.1	+0.4	39.4	39.6	+0.2
Local tensor slices	3	7	17.3	75.3	+58.0	17.1	73.8	+56.7
RoPE rotation	1	1	17.9	17.8	-0.1	17.5	17.4	-0.1
Local tensor concatenation	2	3	37.9	86.5	+48.7	38.6	82.8	+44.2
Rotary weight tiling	1	1	5.7	5.7	+0.0	5.7	5.7	-0.0
Elementwise operations	4	4	60.2	60.0	-0.2	59.0	59.8	+0.7
Rotary transform	1	1	7.9	7.9	-0.0	7.8	8.0	+0.1
Rotary-channel gather	1	1	61.9	61.8	-0.0	61.9	61.8	-0.1
KV dtype conversion	1	2	3.5	7.4	+3.9	3.5	7.2	+3.7
KV-cache writes	1	2	7.8	21.2	+13.4	7.8	21.2	+13.4
Attention (SDPA)	1	2	652.7	491.9	-160.8	16,362.6	14,364.2	-1,998.4
Combine attention heads	1	1	53.8	54.0	+0.2	51.3	50.5	-0.8
Attention output projection	1	1	151.8	152.9	+1.1	150.8	148.2	-2.6
TP reduce-scatter	2	2	237.6	232.9	-4.7	389.0	370.0	-18.9
MLP gate projection	1	1	223.7	223.1	-0.6	221.1	219.8	-1.3
MLP up projection	1	1	155.5	155.9	+0.5	153.1	153.1	+0.0
MLP down projection	1	1	174.5	174.9	+0.4	174.2	174.8	+0.6
TOTAL	41	49	2,412.4	2,372.7	-39.7	18,254.2	16,344.5	-1,909.7

The five projection/MLP matmuls have the same packed row count. Attention, cache writes and slicing operate per request. Counts include all 2 requests. Slices/concatenation copy local rows; norm redistribution changes local memory layout.

The SDPA time includes its internal ring KV exchange. The separate TP all-gather/reduce-scatter operations communicate between tensor-parallel devices. Ordinary ops use the slowest device; collectives use the device average.

Existing global SDPA settings: canonical 8K uses Q blocks of 96, K blocks of 256, one K partition; each 4K call uses Q blocks of 128, K blocks of 256, three K partitions. Both use LoFi. These profiles do not isolate the effect of each setting.

Sliding-window layer: operation comparison

All times are microseconds, summed over the calls in one layer. C = Canonical 1×8K; B = Batch 2×4K.

Operation	Calls C	Calls B	First C μs	First B μs	Extra μs	Last C μs	Last B μs	Extra μs
Norm input redistribution	4	4	27.2	27.5	+0.3	27.1	26.8	-0.3
RMS normalization	7	7	77.3	77.7	+0.4	76.9	76.7	-0.2
Norm output redistribution	4	4	30.3	30.1	-0.2	30.0	30.5	+0.5
TP all-gather	2	2	213.5	212.5	-0.9	205.8	202.4	-3.4
Q/K/V projection	1	1	157.2	157.2	-0.1	158.0	156.1	-1.9
Split Q/K/V heads	1	1	31.2	32.0	+0.8	31.2	32.0	+0.8
RoPE rotation	2	2	44.8	44.9	+0.1	44.5	44.5	-0.0
KV dtype conversion	2	4	9.3	32.1	+22.7	9.3	30.4	+21.0
KV-cache writes	2	4	18.8	37.3	+18.6	18.8	36.5	+17.7
Attention (SDPA)	1	2	356.7	424.7	+68.0	383.5	474.4	+90.9
Combine attention heads	1	1	27.1	27.4	+0.3	26.1	26.4	+0.2
Attention output projection	1	1	101.7	103.2	+1.4	100.3	100.4	+0.2
TP reduce-scatter	2	2	233.0	228.4	-4.6	232.9	220.7	-12.2
Elementwise operations	3	3	55.6	56.1	+0.6	55.0	55.4	+0.4
MLP gate projection	1	1	223.7	223.6	-0.2	222.0	222.0	+0.0
MLP up projection	1	1	156.2	156.1	-0.1	154.3	153.9	-0.4
MLP down projection	1	1	174.9	174.9	-0.0	174.9	174.0	-0.9
Local tensor slices	0	6	0.0	55.9	+55.9	0.0	54.3	+54.3
Local tensor concatenation	0	1	0.0	26.4	+26.4	0.0	24.4	+24.4
TOTAL	36	48	1,938.6	2,128.1	+189.5	1,950.4	2,141.8	+191.3

The five projection/MLP matmuls have the same packed row count. Attention, cache writes and slicing operate per request. Counts include all 2 requests. Slices/concatenation copy local rows; norm redistribution changes local memory layout.

The SDPA time includes its internal ring KV exchange. The separate TP all-gather/reduce-scatter operations communicate between tensor-parallel devices. Ordinary ops use the slowest device; collectives use the device average.

Existing sliding SDPA settings: both paths use Q/K blocks of 128 tokens, one K partition and HiFi2. The batch executes the attention operation twice, once per request.

Global, first: Canonical 1×8K

tt-perf-report per-call table | request tokens [0, 8,192) | kernel sum 2.412 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	7.7	—	48	—	—	—	—	DRAM
2	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
3	Sharded → interleaved	10.1	0.9	48	—	—	—	—	L1 block
4	AllGather	105.5	8.3	36	—	—	—	—	DRAM
5	Matmul 1024 x 5376 x 4608	160.0	0.6	96	282.7	317.2	SLOW	LoFi	DRAM
6	NlpCreateHeads	38.7	0.6	32	—	—	—	—	DRAM
7	LayerNorm	27.4	0.5	120	—	—	—	—	L1
8	LayerNorm	7.9	0.5	32	—	—	—	—	L1
9	Slice	4.7	0.5	120	—	—	—	—	L1
10	Slice	10.9	0.6	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.9	0.6	120	—	—	—	—	L1
12	Concat	34.3	0.5	120	—	—	—	—	L1
13	Slice	1.6	0.6	120	—	—	—	—	L1
14	TilizeWithValPadding	5.7	1.1	1	—	—	—	—	DRAM
15	BinaryNg	4.1	0.5	120	—	—	—	—	L1
16	RotaryEmbedding	7.9	0.5	32	—	—	—	—	L1
17	GatherCodegen	61.9	0.6	120	—	—	—	—	L1
18	Concat	3.6	0.5	120	—	—	—	—	L1
19	Typecast	3.5	0.5	120	—	—	—	—	L1
20	UpdatePaddedKvCache	7.8	0.5	32	—	—	—	—	DRAM_ND_SHARDED
21	RingJointSDPA	652.7	13.0	114	—	—	—	—	DRAM
22	NLPConcatHeads	53.8	2.5	32	—	—	—	—	DRAM
23	Matmul 1024 x 4096 x 5376	151.8	0.5	96	272.9	297.1	SLOW	LoFi	DRAM
24	ReduceScatter	124.0	0.5	36	—	—	—	—	DRAM
25	Interleaved → sharded	7.6	0.5	48	—	—	—	—	DRAM
26	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
27	Sharded → interleaved	5.1	0.9	48	—	—	—	—	L1 block
28	BinaryNg	11.5	0.5	120	—	—	—	—	DRAM
29	Interleaved → sharded	4.3	0.5	48	—	—	—	—	L1
30	LayerNorm	9.7	0.5	48	—	—	—	—	L1 block
31	Sharded → interleaved	10.0	0.8	48	—	—	—	—	L1 block
32	AllGather	106.7	0.6	36	—	—	—	—	DRAM
33	Matmul 1024 x 5376 x 5376	223.7	0.5	96	178.4	264.6	SLOW	LoFi	DRAM
34	Matmul 1024 x 5376 x 5376	155.5	0.6	96	256.7	380.7	FLOP	LoFi	DRAM
35	BinaryNg	32.5	0.6	120	—	—	—	—	L1
36	Matmul 1024 x 5376 x 5376	174.5	0.5	96	228.8	339.3	SLOW	LoFi	L1
37	ReduceScatter	113.6	0.6	36	—	—	—	—	DRAM
38	Interleaved → sharded	7.6	0.5	48	—	—	—	—	DRAM
39	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
40	Sharded → interleaved	5.2	0.9	48	—	—	—	—	L1 block
41	BinaryNg	12.2	0.6	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Global, first: Batch 2×4K

tt-perf-report per-call table | request tokens [0, 4,096) | kernel sum 2.373 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	7.8	—	48	—	—	—	—	DRAM
2	LayerNorm	9.9	0.5	48	—	—	—	—	L1 block
3	Sharded → interleaved	10.3	1.1	48	—	—	—	—	L1 block
4	AllGather	104.4	8.0	36	—	—	—	—	DRAM
5	Matmul 1024 x 5376 x 4608	160.2	0.5	96	282.3	316.8	SLOW	LoFi	DRAM
6	NlpCreateHeads	39.1	0.6	32	—	—	—	—	DRAM
7	LayerNorm	27.7	0.5	120	—	—	—	—	L1
8	LayerNorm	7.9	0.5	32	—	—	—	—	L1
9	Slice	4.7	0.5	120	—	—	—	—	L1
10	Slice	10.8	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.8	0.5	120	—	—	—	—	L1
12	Concat	34.8	0.5	120	—	—	—	—	L1
13	Slice	1.7	0.6	120	—	—	—	—	L1
14	TilizeWithValPadding	5.7	1.0	1	—	—	—	—	DRAM
15	BinaryNg	4.1	0.5	120	—	—	—	—	L1
16	RotaryEmbedding	7.9	0.5	32	—	—	—	—	L1
17	GatherCodegen	61.8	0.5	120	—	—	—	—	L1
18	Concat	3.6	0.5	120	—	—	—	—	L1
19	Slice	25.5	0.5	120	—	—	—	—	DRAM
20	Slice	3.2	0.6	120	—	—	—	—	L1
21	Typecast	3.6	0.6	120	—	—	—	—	DRAM
22	UpdatePaddedKvCache	9.9	0.5	16	—	—	—	—	DRAM_ND_SHARDED
23	RingJointSDPA	243.0	6.8	114	—	—	—	—	DRAM
24	Slice	26.2	0.6	120	—	—	—	—	DRAM
25	Slice	3.3	0.6	120	—	—	—	—	L1
26	Typecast	3.8	0.6	120	—	—	—	—	DRAM
27	UpdatePaddedKvCache	11.3	0.6	16	—	—	—	—	DRAM_ND_SHARDED
28	RingJointSDPA	248.9	0.6	114	—	—	—	—	DRAM
29	Concat	48.1	0.6	120	—	—	—	—	DRAM
30	NLPConcatHeads	54.0	0.6	32	—	—	—	—	DRAM
31	Matmul 1024 x 4096 x 5376	152.9	0.5	96	270.8	294.9	SLOW	LoFi	DRAM
32	ReduceScatter	119.9	0.5	36	—	—	—	—	DRAM
33	Interleaved → sharded	7.6	0.6	48	—	—	—	—	DRAM
34	LayerNorm	9.9	0.5	48	—	—	—	—	L1 block
35	Sharded → interleaved	5.1	0.8	48	—	—	—	—	L1 block
36	BinaryNg	11.4	0.6	120	—	—	—	—	DRAM
37	Interleaved → sharded	4.4	0.5	48	—	—	—	—	L1
38	LayerNorm	9.7	0.5	48	—	—	—	—	L1 block
39	Sharded → interleaved	10.1	0.7	48	—	—	—	—	L1 block
40	AllGather	107.0	0.6	36	—	—	—	—	DRAM
41	Matmul 1024 x 5376 x 5376	223.1	0.5	96	178.9	265.3	SLOW	LoFi	DRAM
42	Matmul 1024 x 5376 x 5376	155.9	0.6	96	255.9	379.6	FLOP	LoFi	DRAM
43	BinaryNg	32.7	0.6	120	—	—	—	—	L1
44	Matmul 1024 x 5376 x 5376	174.9	0.5	96	228.2	338.5	SLOW	LoFi	L1
45	ReduceScatter	113.0	0.6	36	—	—	—	—	DRAM
46	Interleaved → sharded	7.6	0.6	48	—	—	—	—	DRAM
47	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
48	Sharded → interleaved	5.2	0.9	48	—	—	—	—	L1 block
49	BinaryNg	11.7	0.6	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Global, last: Canonical 1×8K

tt-perf-report per-call table | request tokens [253,952, 262,144) | kernel sum 18.254 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	7.7	—	48	—	—	—	—	DRAM
2	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
3	Sharded → interleaved	10.1	0.9	48	—	—	—	—	L1 block
4	AllGather	101.4	8.3	36	—	—	—	—	DRAM
5	Matmul 1024 x 5376 x 4608	158.8	0.6	96	284.8	319.5	SLOW	LoFi	DRAM
6	NlpCreateHeads	39.4	0.6	32	—	—	—	—	DRAM
7	LayerNorm	27.4	0.5	120	—	—	—	—	L1
8	LayerNorm	7.9	0.5	32	—	—	—	—	L1
9	Slice	4.7	0.5	120	—	—	—	—	L1
10	Slice	10.8	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.5	0.5	120	—	—	—	—	L1
12	Concat	35.1	0.5	120	—	—	—	—	L1
13	Slice	1.6	0.6	120	—	—	—	—	L1
14	TilizeWithValPadding	5.7	1.0	1	—	—	—	—	DRAM
15	BinaryNg	4.0	0.5	120	—	—	—	—	L1
16	RotaryEmbedding	7.8	0.5	32	—	—	—	—	L1
17	GatherCodegen	61.9	0.5	120	—	—	—	—	L1
18	Concat	3.6	0.5	120	—	—	—	—	L1
19	Typecast	3.5	0.5	120	—	—	—	—	L1
20	UpdatePaddedKvCache	7.8	0.5	32	—	—	—	—	DRAM_ND_SHARDED
21	RingJointSDPA	16,362.6	13.3	114	—	—	—	—	DRAM
22	NLPConcatHeads	51.3	2.5	32	—	—	—	—	DRAM
23	Matmul 1024 x 4096 x 5376	150.8	0.5	96	274.7	299.1	SLOW	LoFi	DRAM
24	ReduceScatter	277.0	0.5	36	—	—	—	—	DRAM
25	Interleaved → sharded	7.5	0.5	48	—	—	—	—	DRAM
26	LayerNorm	9.9	0.5	48	—	—	—	—	L1 block
27	Sharded → interleaved	4.8	0.8	48	—	—	—	—	L1 block
28	BinaryNg	11.0	0.5	120	—	—	—	—	DRAM
29	Interleaved → sharded	4.3	0.5	48	—	—	—	—	L1
30	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
31	Sharded → interleaved	10.3	0.7	48	—	—	—	—	L1 block
32	AllGather	101.6	0.6	36	—	—	—	—	DRAM
33	Matmul 1024 x 5376 x 5376	221.1	0.5	96	180.5	267.7	SLOW	LoFi	DRAM
34	Matmul 1024 x 5376 x 5376	153.1	0.6	96	260.7	386.6	FLOP	LoFi	DRAM
35	BinaryNg	32.4	0.6	120	—	—	—	—	L1
36	Matmul 1024 x 5376 x 5376	174.2	0.5	96	229.1	339.8	SLOW	LoFi	L1
37	ReduceScatter	112.0	0.6	36	—	—	—	—	DRAM
38	Interleaved → sharded	7.4	0.6	48	—	—	—	—	DRAM
39	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
40	Sharded → interleaved	5.3	0.8	48	—	—	—	—	L1 block
41	BinaryNg	11.6	0.6	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Global, last: Batch 2×4K

tt-perf-report per-call table | request tokens [258,048, 262,144) | kernel sum 16.345 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	7.6	—	48	—	—	—	—	DRAM
2	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
3	Sharded → interleaved	10.4	1.0	48	—	—	—	—	L1 block
4	AllGather	98.1	8.4	36	—	—	—	—	DRAM
5	Matmul 1024 x 5376 x 4608	157.6	0.5	96	286.9	321.9	SLOW	LoFi	DRAM
6	NlpCreateHeads	39.6	0.6	32	—	—	—	—	DRAM
7	LayerNorm	27.1	0.5	120	—	—	—	—	L1
8	LayerNorm	7.9	0.5	32	—	—	—	—	L1
9	Slice	4.6	0.5	120	—	—	—	—	L1
10	Slice	10.9	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.4	0.5	120	—	—	—	—	L1
12	Concat	35.3	0.5	120	—	—	—	—	L1
13	Slice	1.7	0.6	120	—	—	—	—	L1
14	TilizeWithValPadding	5.7	1.0	1	—	—	—	—	DRAM
15	BinaryNg	4.3	0.5	120	—	—	—	—	L1
16	RotaryEmbedding	8.0	0.5	32	—	—	—	—	L1
17	GatherCodegen	61.8	0.5	120	—	—	—	—	L1
18	Concat	3.6	0.5	120	—	—	—	—	L1
19	Slice	24.0	0.5	120	—	—	—	—	DRAM
20	Slice	3.1	0.6	120	—	—	—	—	L1
21	Typecast	3.6	0.6	120	—	—	—	—	DRAM
22	UpdatePaddedKvCache	10.0	0.6	16	—	—	—	—	DRAM_ND_SHARDED
23	RingJointSDPA	7,091.5	6.5	114	—	—	—	—	DRAM
24	Slice	26.1	0.6	120	—	—	—	—	DRAM
25	Slice	3.5	0.6	120	—	—	—	—	L1
26	Typecast	3.6	0.6	120	—	—	—	—	DRAM
27	UpdatePaddedKvCache	11.2	0.6	16	—	—	—	—	DRAM_ND_SHARDED
28	RingJointSDPA	7,272.7	0.5	114	—	—	—	—	DRAM
29	Concat	43.9	0.5	120	—	—	—	—	DRAM
30	NLPConcatHeads	50.5	0.7	32	—	—	—	—	DRAM
31	Matmul 1024 x 4096 x 5376	148.2	0.5	96	279.5	304.3	SLOW	LoFi	DRAM
32	ReduceScatter	257.9	0.5	36	—	—	—	—	DRAM
33	Interleaved → sharded	7.5	0.6	48	—	—	—	—	DRAM
34	LayerNorm	10.0	0.5	48	—	—	—	—	L1 block
35	Sharded → interleaved	5.0	1.0	48	—	—	—	—	L1 block
36	BinaryNg	11.3	0.5	120	—	—	—	—	DRAM
37	Interleaved → sharded	4.3	0.5	48	—	—	—	—	L1
38	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
39	Sharded → interleaved	10.1	1.0	48	—	—	—	—	L1 block
40	AllGather	99.4	0.6	36	—	—	—	—	DRAM
41	Matmul 1024 x 5376 x 5376	219.8	0.6	96	181.6	269.3	SLOW	LoFi	DRAM
42	Matmul 1024 x 5376 x 5376	153.1	0.6	96	260.7	386.6	FLOP	LoFi	DRAM
43	BinaryNg	32.4	0.6	120	—	—	—	—	L1
44	Matmul 1024 x 5376 x 5376	174.8	0.5	96	228.3	338.6	SLOW	LoFi	L1
45	ReduceScatter	112.1	0.5	36	—	—	—	—	DRAM
46	Interleaved → sharded	7.5	0.5	48	—	—	—	—	DRAM
47	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
48	Sharded → interleaved	5.0	1.0	48	—	—	—	—	L1 block
49	BinaryNg	11.7	0.5	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Sliding, first: Canonical 1×8K

tt-perf-report per-call table | request tokens [0, 8,192) | kernel sum 1.939 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	7.7	—	48	—	—	—	—	DRAM
2	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
3	Sharded → interleaved	10.1	1.0	48	—	—	—	—	L1 block
4	AllGather	105.4	8.2	36	—	—	—	—	DRAM
5	Matmul 1024 x 5376 x 4096	157.2	0.6	96	263.4	286.8	SLOW	LoFi	DRAM
6	NlpCreateHeads	31.2	0.6	32	—	—	—	—	DRAM
7	LayerNorm	17.2	0.5	120	—	—	—	—	L1
8	LayerNorm	11.6	0.5	120	—	—	—	—	L1
9	LayerNorm	9.3	0.5	120	—	—	—	—	L1
10	RotaryEmbeddingLlama	26.8	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	18.0	0.5	120	—	—	—	—	L1
12	Typecast	4.7	0.5	120	—	—	—	—	L1
13	UpdatePaddedKvCache	9.2	0.5	120	—	—	—	—	DRAM_ND_SHARDED
14	Typecast	4.6	0.6	120	—	—	—	—	L1
15	UpdatePaddedKvCache	9.6	0.6	120	—	—	—	—	DRAM_ND_SHARDED
16	RingJointSDPA	356.7	9.6	112	—	—	—	—	DRAM
17	NLPConcatHeads	27.1	0.6	32	—	—	—	—	DRAM
18	Matmul 1024 x 2048 x 5376	101.7	0.5	96	257.7	221.7	SLOW	LoFi	DRAM
19	ReduceScatter	118.7	0.6	36	—	—	—	—	DRAM
20	Interleaved → sharded	7.6	0.6	48	—	—	—	—	DRAM
21	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
22	Sharded → interleaved	5.0	0.8	48	—	—	—	—	L1 block
23	BinaryNg	11.4	0.6	120	—	—	—	—	DRAM
24	Interleaved → sharded	4.2	0.5	48	—	—	—	—	L1
25	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
26	Sharded → interleaved	10.1	0.9	48	—	—	—	—	L1 block
27	AllGather	108.0	0.6	36	—	—	—	—	DRAM
28	Matmul 1024 x 5376 x 5376	223.7	0.5	96	178.4	264.6	SLOW	LoFi	DRAM
29	Matmul 1024 x 5376 x 5376	156.2	0.6	96	255.4	378.8	FLOP	LoFi	DRAM
30	BinaryNg	32.0	0.6	120	—	—	—	—	L1
31	Matmul 1024 x 5376 x 5376	174.9	0.5	96	228.1	338.3	SLOW	LoFi	L1
32	ReduceScatter	114.3	11.1	36	—	—	—	—	DRAM
33	Interleaved → sharded	7.7	0.5	48	—	—	—	—	DRAM
34	LayerNorm	9.9	0.5	48	—	—	—	—	L1 block
35	Sharded → interleaved	5.1	0.8	48	—	—	—	—	L1 block
36	BinaryNg	12.1	0.5	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Sliding, first: Batch 2×4K

tt-perf-report per-call table | request tokens [0, 4,096) | kernel sum 2.128 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	7.9	—	48	—	—	—	—	DRAM
2	LayerNorm	9.9	0.5	48	—	—	—	—	L1 block
3	Sharded → interleaved	10.2	0.9	48	—	—	—	—	L1 block
4	AllGather	105.2	8.1	36	—	—	—	—	DRAM
5	Matmul 1024 x 5376 x 4096	157.2	0.6	96	263.5	286.9	SLOW	LoFi	DRAM
6	NlpCreateHeads	32.0	0.6	32	—	—	—	—	DRAM
7	LayerNorm	17.0	0.5	120	—	—	—	—	L1
8	LayerNorm	12.1	0.5	120	—	—	—	—	L1
9	LayerNorm	9.4	0.5	120	—	—	—	—	L1
10	RotaryEmbeddingLlama	26.9	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	18.0	0.6	120	—	—	—	—	L1
12	Slice	17.6	0.6	120	—	—	—	—	DRAM
13	Slice	5.3	0.6	120	—	—	—	—	L1
14	Slice	5.1	0.6	120	—	—	—	—	L1
15	Typecast	7.9	0.6	120	—	—	—	—	DRAM
16	UpdatePaddedKvCache	8.9	0.6	64	—	—	—	—	DRAM_ND_SHARDED
17	Typecast	7.9	0.5	120	—	—	—	—	DRAM
18	UpdatePaddedKvCache	9.3	0.5	64	—	—	—	—	DRAM_ND_SHARDED
19	RingJointSDPA	216.9	0.6	112	—	—	—	—	DRAM
20	Slice	17.4	0.5	120	—	—	—	—	DRAM
21	Slice	5.2	0.6	120	—	—	—	—	L1
22	Slice	5.4	0.6	120	—	—	—	—	L1
23	Typecast	7.8	0.6	120	—	—	—	—	DRAM
24	UpdatePaddedKvCache	9.7	0.5	64	—	—	—	—	DRAM_ND_SHARDED
25	Typecast	8.5	0.5	120	—	—	—	—	DRAM
26	UpdatePaddedKvCache	9.4	0.5	64	—	—	—	—	DRAM_ND_SHARDED
27	RingJointSDPA	207.9	0.5	112	—	—	—	—	DRAM
28	Concat	26.4	0.5	120	—	—	—	—	DRAM
29	NLPConcatHeads	27.4	0.7	32	—	—	—	—	DRAM
30	Matmul 1024 x 2048 x 5376	103.2	0.5	96	254.1	218.6	SLOW	LoFi	DRAM
31	ReduceScatter	114.9	0.5	36	—	—	—	—	DRAM
32	Interleaved → sharded	7.7	0.6	48	—	—	—	—	DRAM
33	LayerNorm	9.9	0.5	48	—	—	—	—	L1 block
34	Sharded → interleaved	4.7	0.9	48	—	—	—	—	L1 block
35	BinaryNg	11.5	0.5	120	—	—	—	—	DRAM
36	Interleaved → sharded	4.3	0.5	48	—	—	—	—	L1
37	LayerNorm	9.7	0.5	48	—	—	—	—	L1 block
38	Sharded → interleaved	10.1	0.6	48	—	—	—	—	L1 block
39	AllGather	107.4	0.6	36	—	—	—	—	DRAM
40	Matmul 1024 x 5376 x 5376	223.6	0.6	96	178.5	264.8	SLOW	LoFi	DRAM
41	Matmul 1024 x 5376 x 5376	156.1	0.6	96	255.6	379.1	FLOP	LoFi	DRAM
42	BinaryNg	32.6	0.6	120	—	—	—	—	L1
43	Matmul 1024 x 5376 x 5376	174.9	0.5	96	228.2	338.4	SLOW	LoFi	L1
44	ReduceScatter	113.5	0.6	36	—	—	—	—	DRAM
45	Interleaved → sharded	7.7	0.6	48	—	—	—	—	DRAM
46	LayerNorm	9.9	0.5	48	—	—	—	—	L1 block
47	Sharded → interleaved	5.2	0.9	48	—	—	—	—	L1 block
48	BinaryNg	12.0	0.5	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Sliding, last: Canonical 1×8K

tt-perf-report per-call table | request tokens [253,952, 262,144) | kernel sum 1.950 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	7.7	—	48	—	—	—	—	DRAM
2	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
3	Sharded → interleaved	10.2	1.0	48	—	—	—	—	L1 block
4	AllGather	101.1	8.4	36	—	—	—	—	DRAM
5	Matmul 1024 x 5376 x 4096	158.0	0.5	96	262.1	285.4	SLOW	LoFi	DRAM
6	NlpCreateHeads	31.2	0.6	32	—	—	—	—	DRAM
7	LayerNorm	16.7	0.5	120	—	—	—	—	L1
8	LayerNorm	11.6	0.5	120	—	—	—	—	L1
9	LayerNorm	9.2	0.5	120	—	—	—	—	L1
10	RotaryEmbeddingLlama	26.7	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.9	0.5	120	—	—	—	—	L1
12	Typecast	4.7	0.5	120	—	—	—	—	L1
13	UpdatePaddedKvCache	9.4	0.5	120	—	—	—	—	DRAM_ND_SHARDED
14	Typecast	4.6	0.6	120	—	—	—	—	L1
15	UpdatePaddedKvCache	9.4	0.6	120	—	—	—	—	DRAM_ND_SHARDED
16	RingJointSDPA	383.5	9.4	112	—	—	—	—	DRAM
17	NLPConcatHeads	26.1	0.6	32	—	—	—	—	DRAM
18	Matmul 1024 x 2048 x 5376	100.3	0.5	96	261.4	224.9	SLOW	LoFi	DRAM
19	ReduceScatter	117.8	0.5	36	—	—	—	—	DRAM
20	Interleaved → sharded	7.6	0.6	48	—	—	—	—	DRAM
21	LayerNorm	9.9	0.5	48	—	—	—	—	L1 block
22	Sharded → interleaved	4.9	1.0	48	—	—	—	—	L1 block
23	BinaryNg	11.2	0.6	120	—	—	—	—	DRAM
24	Interleaved → sharded	4.3	0.5	48	—	—	—	—	L1
25	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
26	Sharded → interleaved	10.1	0.7	48	—	—	—	—	L1 block
27	AllGather	104.6	0.6	36	—	—	—	—	DRAM
28	Matmul 1024 x 5376 x 5376	222.0	0.6	96	179.8	266.7	SLOW	LoFi	DRAM
29	Matmul 1024 x 5376 x 5376	154.3	0.6	96	258.7	383.6	FLOP	LoFi	DRAM
30	BinaryNg	32.1	0.6	120	—	—	—	—	L1
31	Matmul 1024 x 5376 x 5376	174.9	0.5	96	228.2	338.4	SLOW	LoFi	L1
32	ReduceScatter	115.0	11.0	36	—	—	—	—	DRAM
33	Interleaved → sharded	7.5	0.5	48	—	—	—	—	DRAM
34	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
35	Sharded → interleaved	4.8	0.8	48	—	—	—	—	L1 block
36	BinaryNg	11.7	0.6	120	—	—	—	—	L1

Source: tt-perf-report 1.4.1, main commit cb9407747a28. Rows follow execution order. The initial idle gap is excluded. Device columns summarize the 32 devices; they are not added together.

DRAM bandwidth, compute throughput and Bound are the tool's estimates. Blank entries mean the tool has no model for that operation; they do not mean zero cost. CSV/text reports retain the complete tool output and advice.

Sliding, last: Batch 2×4K

tt-perf-report per-call table | request tokens [258,048, 262,144) | kernel sum 2.142 ms

#	Operation / matmul M × K × N	Device μs	Gap μs	Cores	DRAM GB/s	Compute TFLOP/s	Bound	Math	Input 0 storage
1	Interleaved → sharded	7.6	—	48	—	—	—	—	DRAM
2	LayerNorm	9.9	0.5	48	—	—	—	—	L1 block
3	Sharded → interleaved	10.3	0.9	48	—	—	—	—	L1 block
4	AllGather	100.0	8.8	36	—	—	—	—	DRAM
5	Matmul 1024 x 5376 x 4096	156.1	0.6	96	265.3	288.8	SLOW	LoFi	DRAM
6	NlpCreateHeads	32.0	0.6	32	—	—	—	—	DRAM
7	LayerNorm	16.5	0.5	120	—	—	—	—	L1
8	LayerNorm	11.6	0.5	120	—	—	—	—	L1
9	LayerNorm	9.3	0.5	120	—	—	—	—	L1
10	RotaryEmbeddingLlama	26.7	0.5	120	—	—	—	—	L1
11	RotaryEmbeddingLlama	17.8	0.5	120	—	—	—	—	L1
12	Slice	16.9	0.5	120	—	—	—	—	DRAM
13	Slice	5.3	0.6	120	—	—	—	—	L1
14	Slice	5.1	0.6	120	—	—	—	—	L1
15	Typecast	7.3	0.6	120	—	—	—	—	DRAM
16	UpdatePaddedKvCache	8.9	0.5	64	—	—	—	—	DRAM_ND_SHARDED
17	Typecast	7.7	0.6	120	—	—	—	—	DRAM
18	UpdatePaddedKvCache	9.7	0.5	64	—	—	—	—	DRAM_ND_SHARDED
19	RingJointSDPA	239.3	0.6	112	—	—	—	—	DRAM
20	Slice	16.7	0.6	120	—	—	—	—	DRAM
21	Slice	5.1	0.6	120	—	—	—	—	L1
22	Slice	5.2	0.6	120	—	—	—	—	L1
23	Typecast	8.0	0.6	120	—	—	—	—	DRAM
24	UpdatePaddedKvCache	9.1	0.5	64	—	—	—	—	DRAM_ND_SHARDED
25	Typecast	7.4	0.5	120	—	—	—	—	DRAM
26	UpdatePaddedKvCache	8.8	0.5	64	—	—	—	—	DRAM_ND_SHARDED
27	RingJointSDPA	235.1	0.5	112	—	—	—	—	DRAM
28	Concat	24.4	0.5	120	—	—	—	—	DRAM
29	NLPConcatHeads	26.4	0.7	32	—	—	—	—	DRAM
30	Matmul 1024 x 2048 x 5376	100.4	0.5	96	261.0	224.5	SLOW	LoFi	DRAM
31	ReduceScatter	109.8	0.6	36	—	—	—	—	DRAM
32	Interleaved → sharded	7.5	0.5	48	—	—	—	—	DRAM
33	LayerNorm	9.8	0.5	48	—	—	—	—	L1 block
34	Sharded → interleaved	5.1	0.8	48	—	—	—	—	L1 block
35	BinaryNg	11.2	0.5	120	—	—	—	—	DRAM
36	Interleaved → sharded	4.2	0.5	48	—	—	—	—	L1
37	LayerNorm	9.7	0.5	48	—	—	—	—	L1 block
38	Sharded → interleaved	10.2	0.6	48	—	—	—	—	L1 block
39	AllGather	102.4	0.6	36	—	—	—	—	DRAM
40	Matmul 1024 x 5376 x 5376	222.0	0.6	96	179.8	266.6	SLOW	LoFi	DRAM
41	Matmul 1024 x 5376 x 5376	153.9	0.6	96	259.3	384.5	FLOP	LoFi	DRAM
42	BinaryNg	32.2	0.6	120	—	—	—	—	L1
43	Matmul 1024 x 5376 x 5376	174.0	0.5	96	229.4	340.2	SLOW	LoFi	L1
44	ReduceScatter	110.8	0.6	36	—	—	—	—	DRAM
45	Interleaved → sharded	7.5	0.5	48	—	—	—	—	DRAM
46	LayerNorm	9.9	0.5	48	—	—	—	—	L1 block
47	Sharded → interleaved	4.9	0.8	48	—	—	—	—	L1 block
48	BinaryNg	11.9	0.6	120	—	—	—	—	L1

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